Generation by panels compatible with a partial family

Human Compression project

Let $\mathcal{P}_B$ be the acquired complete panel set for a fixed $B \leq G$. Write $P \sim Q$ when the acquired rank-two test accepts the distinct panels P, Q. Every family under consideration has mutually compatible panels. This is a necessary condition; no converse is used. For a selected family $T \subseteq \mathcal{P}_B$, put

$$C(T) = T \cup \{P \in \mathcal{P}_B \setminus T : P \sim Q \text{ for every } Q \in T\}, \qquad J(T) = \langle B, P \mid P \in C(T) \rangle.$$

Lemma 1. *If $J(T) < G$, no admissible family containing T generates G. The exclusion is invariant under conjugation by $N_G(B)$ and under reordering the selected panels.*

Proof. Any additional panel in an admissible family must be compatible with each member of T, and hence lies in $C(T)$. The whole family therefore generates a subgroup of $J(T)$. Properness proves the first assertion. The acquired rank-two conditions are properties of actual subgroup intersections, joins, maximality and coset incidence; conjugation preserves each of them. Thus conjugation sends $C(T)$ onto the compatible set of the conjugate family, and sends $J(T)$ to its conjugate. Reordering T changes neither set. □

Consequently this exclusion can be inserted into the acquired maximal augmentation search for families generating G. For any such final family F, the acquired augmentation theorem supplies a chain of admissible prefixes leading to a conjugate and reordering of F. For each prefix T on that chain, $F \subseteq C(T)$ and therefore $J(T) = G$. No required chain is removed. Equivariance ensures that replacing a prefix by the representative already chosen by the normalizer and type action does not alter this conclusion. The original acceptance tests and the original equivalence test still apply; this additional necessary condition accepts no new family.

One may start from an already complete saved level of the old search. Such a level may contain prefixes subsequently ruled out by this bound. They can be removed or skipped by the same argument. Earlier complete full-group families remain valid. Later level sizes describe the search with this exclusion, rather than the enumeration of all proper generated families. The distinction is retained in the execution records.

The computation forms the symmetric compatibility matrix using the unchanged rank-two test, then forms each actual compatible panel join. For a rejected prefix it records that prefix, its complete compatible index set and explicit generators for the proper join. Its correctness depends on the acquired complete panel set, complete earlier level when resuming, and exact finite-group operations. This is the previous generation bound applied after a necessary local restriction; no new classification or whole-source claim is made by the bound alone.